

What Is a Function?

A set of ordered pairs is a **relation**.

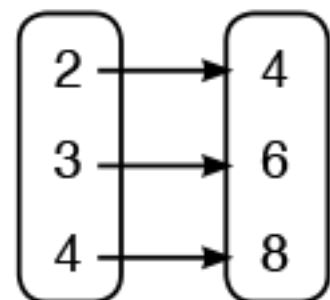
$$\{(2, 4), (3, 6), (4, 8)\} \leftarrow$$

A set is written in braces.

The first member in each ordered pair is part of the **domain**. In this set, 2, 3, and 4 are in the domain. The second member is part of the **range**. The range consists of 4, 6, and 8.

A relation is a **function** when each element of the domain is paired with *exactly one* element of the range.

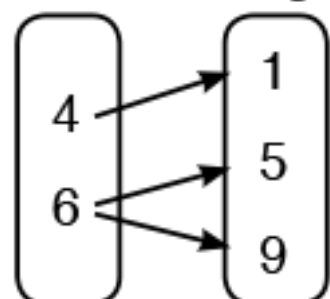
domain range



function

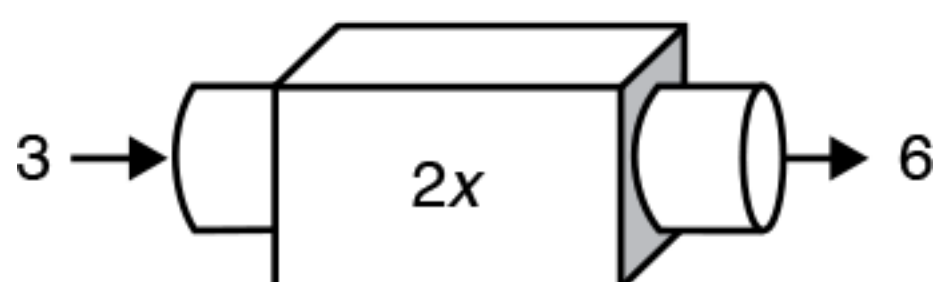
The relation below is *not* a function because 6 is paired with two different elements in the range.

domain range



not a function

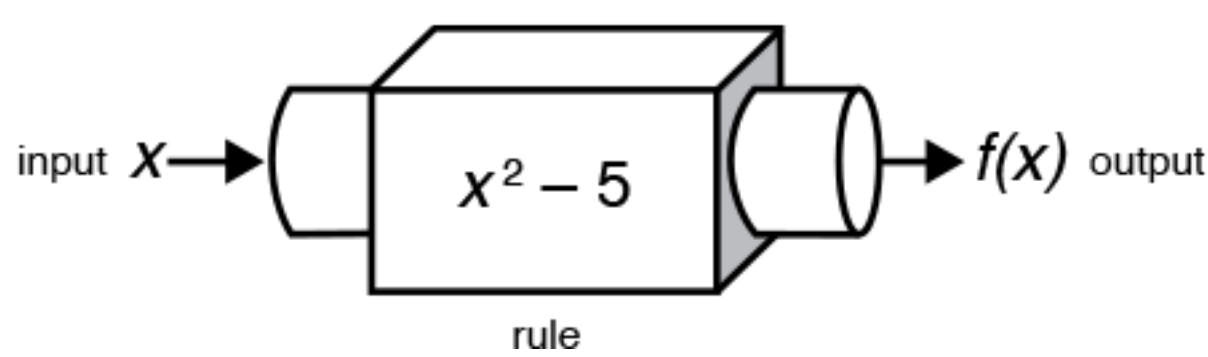
A function is like a machine that applies a rule to each member of the domain. A member of the domain enters the machine. The rule is applied, and a member of the range exits the machine. Here, 3 becomes 6 when the rule “multiply by 2” is applied.



Functions use a special system of notation. The expression $f(x)$ is read “ f of x .” It means “the function of x .” A function states the rule to apply to any value of x .

Note: The letters f , g , and h are usually reserved for the naming of functions.

An example of a function is $f(x) = x^2 - 5$. It says that for every value of x , the function of x is equal to x squared minus 5.



Example

Hide

For $f(x) = x^2 - 5$, find $f(-2)$ and $f(4)$.

Find $f(-2)$.

Substitute -2 for x in the function and solve.

$$\begin{aligned} f(-2) &= (-2)^2 - 5 \\ &= 4 - 5 \\ &= -1 \end{aligned}$$

Find $f(4)$.

Substitute 4 for x in the function and solve.

$$\begin{aligned} f(4) &= 4^2 - 5 \\ &= 16 - 5 \\ &= 11 \end{aligned}$$

$$f(-2) = -1 \text{ and } f(4) = 11$$

Key Terms

 domain

 intervals

 relation

 function

 range

vertical line test